will be attributed to individual SROs[397] in the consolidated market data products made available by competing consolidators. The Commission is therefore adopting a modified version of the definition of "depth of book data."

First, the revised definition of depth of book data will not include the first clause of the proposed definition, which referred to "all quotation sizes at each national securities exchange, aggregated at each price at which there is a bid or offer that is lower than the best bid down to the protected bid and higher than the best offer up to the protected offer." As discussed above, the Commission is not adopting the proposed amendments to the definition of protected bid or protected offer, which would have required such quotations to be "of at least 100 shares."[398] As a result, as is the case today, in the vast majority of cases the best quotes and the protected quotes will be the same.[399] Therefore, the definition of depth of book data does not need to include quotation interest between the best and protected quotes. This modification also addresses commenters' concerns that this aspect of the proposed definition would have included an

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397 As discussed below, an indicator of the national securities exchange or national securities association on which the liquidity at a depth of book price level resides will be included in the adopted definition of depth of book data.

398 See supra Section II.E.2.

399 Among other things, a protected bid or offer must be an "automated quotation," which means "a quotation displayed by a trading center that: (i) Permits an incoming order to be marked as immediate-or-cancel; (ii) Immediately and automatically executes an order marked as immediate-or-cancel against the displayed quotation up to its full size; (iii) Immediately and automatically cancels any unexecuted portion of an order marked as immediate-or-cancel without routing the order elsewhere; (iv) Immediately and automatically transmits a response to the sender of an order marked as immediate-or-cancel indicating the action taken with respect to such order; and (v) Immediately and automatically displays information that updates the displayed quotation to reflect any change to its material terms." Rules 600(b)(6), (70) of Regulation NMS, 17 CFR 242.600(b)(6), (70). Therefore, a "manual quotation" that is not automated—also known as a "slow quote"—can sometimes be the best quotation without being a protected quotation. Rule 600(b)(45) of Regulation NMS, 17 CFR 242.600(b)(45).

excessive amount of information in core data that would be difficult to calculate and consume as prices fluctuate throughout the trading day.

Second, in response to comments suggesting that the proposed definition of depth of book data would require five price levels from each exchange to be included in core data, the Commission is modifying the second clause of the proposed definition so that it refers to the next five prices at which there is a bid (offer) that is lower (higher) than the *national best bid (offer)* rather than five price levels from the *protected bid or offer*. These changes specify that the starting points for the "next 5 prices" are the highest priced bid and lowest priced offer on *any* exchange (i.e., the national best bid and national best offer) rather than on *each* exchange. Thus, the modified definition specifies that depth of book data includes five levels of aggregated quotation sizes; it does not include more than five levels to account for differences in the highest priced bid or lowest priced offer on each exchange.

The following example illustrates the price levels that are included in the definition of depth of book data.

Example 1: Bids for XYZ, Inc. on Three Exchanges

	Exchange A	Exchange B	Exchange C
Best bid	100 shares at $25.02	100 shares at $25.01	100 shares at $25.00
DOB 1	100 shares at $25.01	100 shares at $25.00	100 shares at $24.99
DOB 2	100 shares at $25.00	100 shares at $24.99	100 shares at $24.97
DOB 3	100 shares at $24.99	100 shares at $24.98	100 shares at $24.95
DOB 4	100 shares at $24.98	100 shares at $24.96	100 shares at $24.92
DOB 5	100 shares at $24.97	100 shares at $24.95	100 shares at $24.90

Note: Prices in shaded areas would not be included in core data pursuant to the definition of depth of book data as amended.

In this example, depth of book data would include aggregate quotation sizes at each price level at and between $25.01[400] and $24.97 because the starting point for the "next five prices" is $25.02 (the best bid on Exchange A and the national best bid) rather than $25.01 (the best bid on Exchange B) or $25.00 (the best bid on Exchange C). Depth of book data would not include the interest at or below $24.96, even if that interest is within the top five levels of any given exchange, because it is not within the first five price levels from the national best bid. A competing consolidator could provide interest beyond the first five levels of depth across exchanges but would have to acquire such data on a proprietary basis from the exchanges.[401]

The Commission believes that revising the definition of depth of book data so that the included price levels are the five prices below the national best bid and above the national best offer, rather than the five prices above and below the best quotes on each exchange, is appropriate because it limits the amount of information that competing consolidators will process and display,[402] which mitigates concerns raised by some commenters about adverse consequences—such as unnecessary complexity or increased processing demands and latency—

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400 The Commission notes that, in this example, the interest at $25.02 would also be included in core data because the NBBO is one of the elements of core data.

401 As explained below, Rule 614(d) requires competing consolidators to calculate, generate, and make available to subscribers a consolidated market data product, which can contain all elements of consolidated market data or a subset thereof, but competing consolidators are permitted to offer custom products containing more information as well. Competing consolidators would compensate SROs for the data necessary to generate consolidated market data through fees established by the effective national market system plan(s) and would compensate SROs for data that goes beyond consolidated market data on a proprietary basis pursuant to individual SRO fee schedules. See infra note 1132.

402 See id.

that could result from a broader interpretation of the proposed definition.[403] In addition, the Commission believes, consistent with the views expressed by various commenters, that a broad array of market participants could use five levels of depth of book data away from the national best bid and national best offer to trade in an informed and effective manner.[404]

Third, the Commission is modifying the definition of depth of book data to specify that, in addition to quotation sizes at the first five price levels from the NBBO, the aggregate size at each included price level shall be attributed to each exchange on which the interest is available. Although the proposed definition of depth of book data referred to quotation sizes "at each national securities exchange" in order to require the price and size information in depth of book data to be associated with the source exchange, the Commission believes that specifying that depth of book data includes the aggregate[405] quotation size available at each price at each

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403 See Proposing Release, 85 FR at 16753 (stating that defining depth of book data to include a finite number of price levels would limit "processing demand on systems" and avoid "excessive message traffic or complexity").

404 See Proposing Release, 85 FR at 16753 (stating that depth of book data should "enhance[] the utility of proposed core data for a wide range of market participants" rather than "supplanting the proprietary depth offerings of the exchanges that contain additional content and that may be more appropriate for certain market participants or more specialized use cases"). Market participants in need of more depth than included in the definition of depth of book data could purchase a custom depth product separately from exchanges or through a competing consolidator.

405 For example, if the national best bid is $25.10, Exchanges A and B have 100-share bids at $25.09, and Exchange C has two 100 share bids at $25.09, the competing consolidator would disseminate: 100 shares at Exchange A, 100 shares at Exchange B, 200 shares at Exchange C. The Commission believes that aggregating quotation sizes at each price level at each exchange will limit the number of messages included in depth of book data as compared to an order by order approach, reducing potential concerns about processing demands on systems, latency, and operational costs. See infra Section II.F.2(d).

national securities exchange and national securities association would clarify where the liquidity resides.[406] For instance, with respect to Example 1, depth of book data would include:

$25.01 (100 on Exchange A, 100 on Exchange B)
$25.00 (100 on Exchange A, 100 on Exchange B, 100 on Exchange C)
$24.99 (100 on Exchange A, 100 on Exchange B, 100 on Exchange C)
$24.98 (100 on Exchange A, 100 on Exchange B)
$24.97 (100 on Exchange A, 100 on Exchange C)

The Commission believes that attributing the quotation size at each price to its source national securities exchange or association can, in many circumstances, be essential to achieving the benefits of including depth of book data in consolidated market data. Improved placement of liquidity-taking and liquidity providing orders, for example,[407] requires knowledge of the exchange at which the liquidity resides so that market participants can direct orders to that exchange.

(c) Expand Definition to FINRA's Alternative Display Facility

The proposed definition of depth of book data referred to quotations only on national securities exchanges, but the Commission solicited comment on whether to include the depth of book quotations of national securities associations to account for the possibility of quotes being

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406 See infra Section II.F.2(c) for an explanation of why the Commission is revising the definition of depth of book data to include quotations on a facility of a national securities association.

407 Proposing Release, 85 FR at 16754.

reported to FINRA's Alternative Display Facility ("ADF") in the future.[408] One commenter stated that while the ADF does not currently have quoting participants, it is an actively maintained FINRA facility and could readily add quoting participants in the future.[409] This commenter recommended including future potential ADF quotations by modifying the definition of depth of book data to refer to all quotation sizes at each national securities exchange and "on a facility of a national securities association."[410]

The Commission agrees and is modifying the proposed definition of depth of book data so that it refers to quotation sizes on a facility of a national securities association as well as quotation sizes on a national securities exchange. The Commission agrees with the commenter that any depth of book quotation activity displayed through the ADF or similar facilities is comparable to the exchange depth of book data that would be included as core data under the proposal and that it should be made available on the same terms as exchange depth of book data. To provide market participants with a more complete view of the liquidity that may be available at depth of book price levels, the Commission is modifying the proposed depth of book data definition so that any future ADF depth of book quotation data—or data from similar national securities association facilities that may be developed—would be included in core data without the need for additional Commission rulemaking. In addition, by modifying the definition to refer generally to a facility of a national securities association, rather than specifically to FINRA's

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408 Proposing Release, 85 FR at 16756.

409 FINRA Letter at 12–13.

410 Id. The commenter recommended this particular formulation, rather than a direct reference to the ADF, "to account for the possibility that other quotation facilities may be developed in the future." Id. at 13.

ADF, the definition will include any potential national securities association depth of book quotations.

(d) Include Full Depth of Book and Order by Order Data

Some commenters suggested including full depth of book data in core data rather than five levels of depth of data.[411] One commenter stated that the proposal could be "counterproductive and confusing" and recommended adding full depth of book data, including order-by-order data across all price levels.[412] Another commenter recommended including complete, order-by-order depth of book data in core data, explaining that "the existence of proprietary data feeds alongside a public tape creates incentives which are incompatible with promoting fair and orderly markets."[413] One commenter agreed with the proposed five-levels of depth of book data but stated that, while not necessary for all market participants, providing full depth of book would not tax systems much more than providing five levels, since ninety percent of "quote changes" occur at the top five price levels.[414] However, another commenter stated that providing complete depth of book data is not necessary at this time, explaining that additional data results in increased data processing, latency, and complexity that could impair the usability of the data.[415] Finally, one commenter suggested that the Commission should replicate the full depth of book curve to help subscribers of consolidated data better understand the supply and demand imbalance of liquidity in real time.[416]

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411 See MEMX Letter at 5; BlackRock Letter at 2.

412 See MEMX Letter at 5.

413 See BlackRock Letter at 2.

414 See IntelligentCross Letter at 4.

415 See Clearpool Letter at 14.

416 See Data Boiler Letter I at 19.

The Commission does not believe that the definition of depth of book data should be expanded to include complete, order-by-order depth of book at all price levels. As explained in the Proposing Release, the expansion of NMS information generally, and the inclusion of depth of book data in core data specifically, was intended to provide additional information that would be useful to a broad cross-section of market participants and to reduce information asymmetries between users of proprietary data and users of SIP data.[417] However, the Commission explained that these objectives must be balanced against the risk of excessive complexity, message traffic, processing demand on systems, and associated operational costs that might result from the inclusion of more complete depth of book information.[418] The Commission recognized that some market participants may need more granular and expansive data, such as the proprietary depth of book products offered by many exchanges, for certain use cases.[419] Including complete depth of book data in core data would go beyond the needs of a wide array of market participants or standard use cases for depth of book data in trading, and could result in additional operational costs and latency because of increased message traffic with order by order data at all price levels. Therefore, consistent with the views of several commenters,[420] the Commission continues to believe that while five levels of depth of book data would significantly enhance the usability of core data for many market participants, including complete depth of book data in core data could impair the usability of core data for many subscribers.

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417 Proposing Release, 85 FR at 16734, 53.

418 Id. at 16753.

419 Id.

420 See, e.g., supra note 388.

Moreover, while the Commission has considered the view expressed by one commenter that providing depth of book data at all price levels would not be overly burdensome and would not tax systems much more than providing five levels, this commenter also acknowledged that full depth of book data is not necessary at this time for all market participants.[421] The Commission shares some commenters' concern about the processing and latency ramifications of including complete depth of book data and therefore is not adopting a definition of depth of book data that goes beyond five levels.[422]

(e) Odd-Lots at Depth and Determination of Five Price Levels

One commenter recommended that "depth-of-book quotations aggregated at each of the first five price levels where a displayed order is available to trade … regardless of the associated size displayed at those prices" be disseminated.[423] This commenter requested that the Commission clarify, preferably with an example, how odd-lots would be aggregated at depth of book price levels, stating that the proposed definition of core data requires odd-lots to be aggregated across prices and disseminated at the least aggressive price for purposes of depth of book data, while the proposed definition of depth of book data provides that the required five

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421 See IntelligentCross Letter at 4.

422 Similarly, in response to the comment regarding the full depth of book curve, see supra note 416 and accompanying text, while full order book shape patterns may contain valuable information for certain sophisticated computerized models, the Commission, as discussed above, is concerned that mandating the inclusion of the entire depth of book across all national securities exchanges would have significant processing, latency, and cost ramifications.

423 See Cboe Letter at 15.

price levels are determined by the presence of a "bid" or "offer," which by definition, implies a round lot.[424]

The Commission does not agree that all odd-lots within the first five price levels of the NBBO should be displayed in core data or that the five price levels included in depth of book data should be determined by the presence of an odd-lot quotation at those price levels. First, as explained in detail above, the inclusion of odd-lot quotations in core data must be reasonably calibrated to include the information that is most relevant to investors and other market participants.[425] While the Commission believes that information on the most attractively priced individual odd-lots—namely, those priced at or better than the NBBO—should be included in core data, the inclusion of information regarding individual odd-lots at inferior prices is not warranted because these quotations do not represent direct and immediate opportunities for price improvement.[426] Second, the magnitude of the quotation size available at a particular price level should factor in to whether that price level counts as one of the five price levels that are included in core data. Otherwise, particularly in cases where liquidity is widely dispersed over a number of price levels, as is the case with many illiquid stocks, orders of insignificant notional value could "take up" price levels and prevent the dissemination of more significant interest at price levels further away. For example, if the NBB for stock A is $25.15, and there are 1 share bids at

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424 See id. at 16–17; see also SIFMA Letter at 7 (requesting clarification regarding how odd-lots will be aggregated at depth of book price levels).

425 See supra Section II.C.2(b).

426 See also infra Section II.F.2(h) (explaining the relevance of depth of book data to best execution analyses); Proposing Release, 85 FR at 16754 (explaining how depth of book data can enable market participants to trade in a more informed and effective manner).